

PyGameMaker — Tutorial 15: Fruit Fusion

How to Use This Handout

This handout goes with the in-app tutorial **Help > Tutorials > Fruit Fusion: Catch and Merge!** (5 pages, about 20-25 minutes). Keep the tutorial open on one half of your screen. Tick each box when the step is done, and check the **You should see** line at the end of every phase. If something does not match, look at **Stuck?** before you call your teacher.

What You Will Make

A basket that always holds one fruit, starting with a cherry. Cherries, strawberries and oranges fall from the sky. Catch a fruit that matches what you are holding and it fuses into the next, bigger fruit. Fuse all the way up to a watermelon and you win!

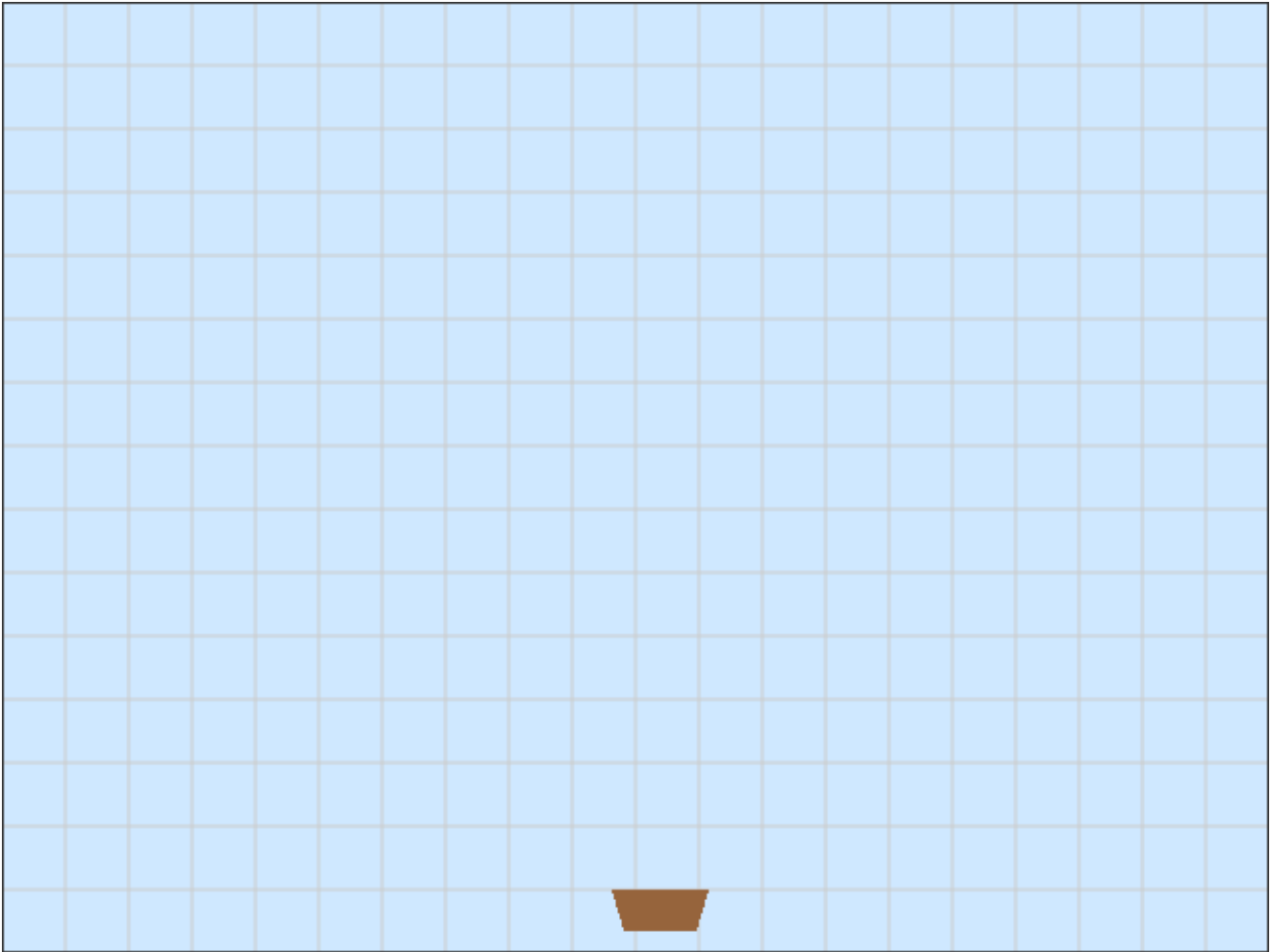
YOU WIN! You made a watermelon!

Press SPACE to play again

Phase 1: Moving Basket

- ☐ **1.** Make a new project FruitFusion.
- ☐ **2.** Create a sprite spr_basket_empty (56x40, a brown bowl shape) and an object obj_player with that sprite.
- ☐ **3.** Create a room room_main and place obj_player near the bottom center.
- ☐ **4.** In obj_player: **Keyboard: Left Arrow (held)** with *Set horizontal speed to -5*; **Right Arrow (held)** with *5*; **Keyboard: No key** with *Stop movement*.

You should see: press **F5**. Left and Right move the basket, and it stops when you let go.



Phase 2: First Fusion

- ☐ **5.** Create a sprite `spr_fruit_cherry` (20x20, a red circle) and an object `obj_fruit_cherry` with a **Create** event (*Set vertical speed 3*) and an **Outside Room** event (*Destroy this instance*).
- ☐ **6.** Create `obj_spawn_cherry` (no sprite) with a **Create** event (*Set alarm 0 to 90*) and an **Alarm 0** event (*Create instance of `obj_fruit_cherry` at a random x, y 0*, then *Set alarm 0 to 90* again). Place one in `room_main`.
- ☐ **7.** Change `obj_player`'s sprite to a new sprite `spr_basket_cherry` (a bowl with a cherry in it) — the basket now starts out already holding a cherry.
- ☐ **8.** In `obj_player`: **Create** event sets *score to 0* and *variable held_level to 1*; **Draw** event does *Draw score at x 10, y 10*.
- ☐ **9.** Add **Collision with obj_fruit_cherry**: *If held_level equals 1 then* [*set held_level to 2*, *add 10 to score*, *set sprite to `spr_basket_strawberry`*] *Otherwise* [*add 1 to score*]; then *Destroy other instance*.

You should see: press **F5**. Catch a falling cherry — the basket turns into a strawberry and the score jumps by 10.

Score: 10



Phase 3: Second Fusion

This phase repeats the exact same recipe as Phase 2, with "cherry" changed to "strawberry" and the numbers changed from 1/2 to 2/3.

- ☐ **10.** Create a sprite `spr_fruit_strawberry` (26x26, a pink-red circle) and an object `obj_fruit_strawberry`, built exactly like `obj_fruit_cherry`.
- ☐ **11.** Create `obj_spawn_strawberry` (alarm 0 to 120 this time) and place one in `room_main`.
- ☐ **12.** Add **Collision with `obj_fruit_strawberry`** on `obj_player`: *If `held_level` equals 2 then* [*set `held_level` to 3*, *add 20 to score*, *set sprite to `spr_basket_orange`*] *Otherwise* [*add 1 to score*]; then *Destroy other instance*.

You should see: press **F5**. Catch a cherry, then a strawberry — the basket ends up an orange, having scored 30 points.

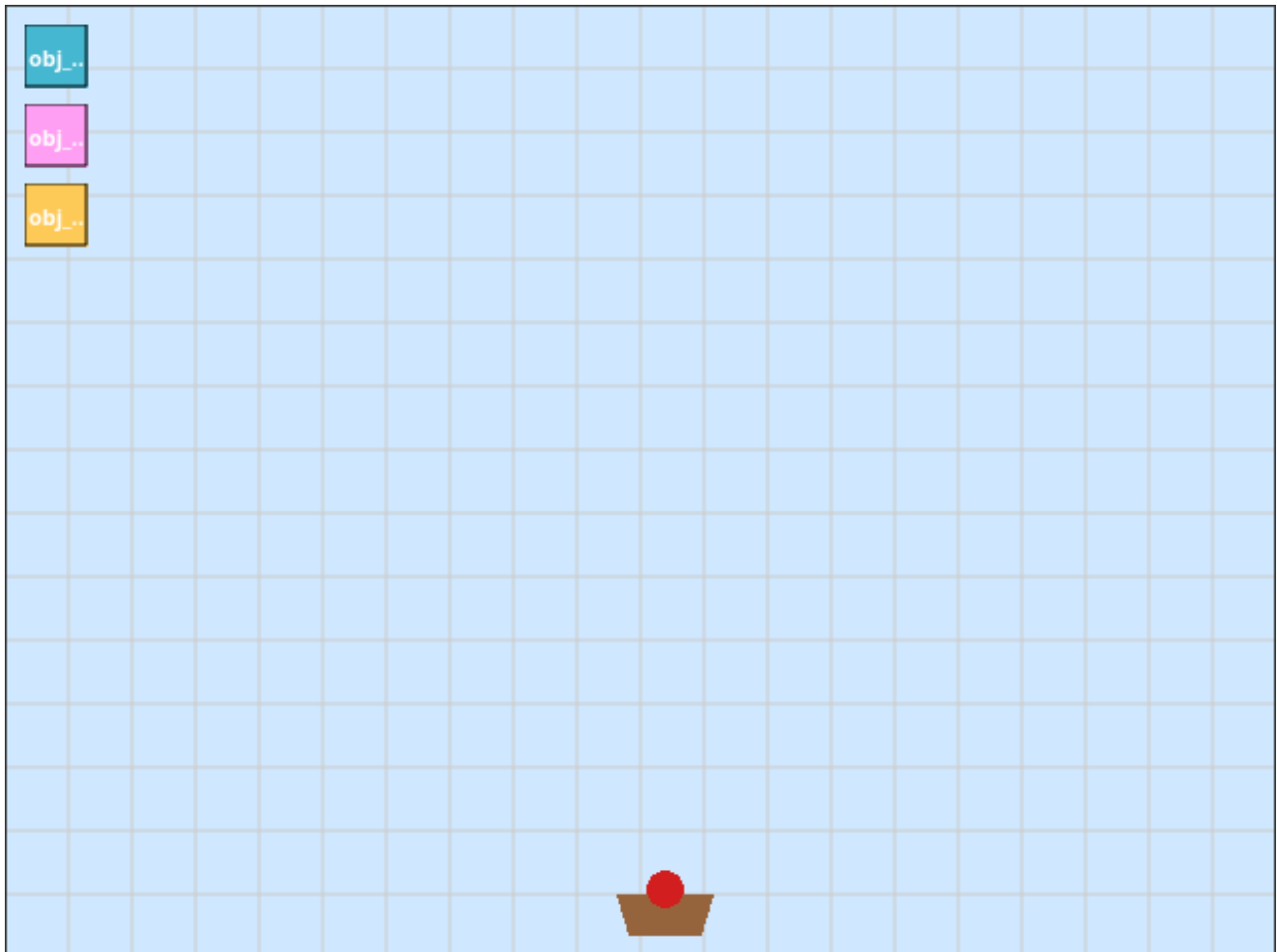
Score: 30



Phase 4: Winning

- ☐ **13.** Create a sprite `spr_fruit_orange` (32x32, an orange circle) and an object `obj_fruit_orange`, built exactly like the others. There is no falling watermelon sprite or spawner — a watermelon is only ever something your basket becomes.
- ☐ **14.** Create `obj_spawn_orange` (alarm 0 to 150) and place one in `room_main`.
- ☐ **15.** Create `obj_win_text`. **Draw** event: `*Draw text* "YOU WIN! You made a watermelon!"` at x 120, y 220 and `"Press SPACE to play again"` at x 190, y 260. **Key press: space:** `*Restart game*`.
- ☐ **16.** Create `room_win` with a bright background color, and place one `obj_win_text` instance in it.
- ☐ **17.** Add **Collision with `obj_fruit_orange`** on `obj_player`: `*If held_level equals 3 then* [*set held_level to 4*, *add 50 to score*, *set sprite to spr_basket_watermelon*, *Go to room room_win*] *Otherwise* [*add 1 to score*]; then *Destroy other instance*`.

You should see: press **F5**. Chain all three fusions and the YOU WIN! room appears. SPACE starts a new game.



Stuck?

What is wrong	Most likely cause	What to do
The basket never changes picture	set_sprite is missing from the *If* branch, or it points at the wrong sprite name	Check the exact sprite name in *Set sprite to*
Catching a matching fruit only gives 1 point	held_level is not actually equal to the number the *If* checks, or the condition uses the wrong operator	Check *If held_level equals* uses the right number for that phase (1, then 2, then 3)
The basket is stuck as a cherry forever	The merge's *Otherwise* branch accidentally changed held_level too	Only the *If* branch should touch held_level and the sprite
Fruit piles up and never disappears	The fruit's Outside Room event is missing	Add *Destroy this instance* in Outside Room
No fruit ever falls	The spawner object has no instance placed in room_main, or its alarm was never set in Create	Place one spawner instance; check *Set alarm 0* is in Create , not somewhere else
Catching an orange does not win	The *Go to room room_win* action is in the wrong place, or the room name is misspelled	It must be inside the *If held_level equals 3* branch, spelled exactly room_win

